A Survey of Inverse Dynamics Models

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I. Related Work

World models learn environment dynamics for prediction and planning, ranging from latent dynamics models to large-scale video-based models trained on diverse sequential data, and offer a scalable framework for policy evaluation and optimization [1], [2]. To improve robustness under limited action supervision, recent works incorporate verification and inverse dynamics constraints to enforce action-consistent predictions and enable self-improvement [3]. Inverse dynamics models (IDMs), which infer actions from state transitions, they can be broadly categorized into explicit and implicit approaches.

A. Implicit Inverse Dynamics Modeling

Implicit inverse dynamics modeling captures the relationship between state transitions and actions without introducing an explicit action prediction module. Instead, actions are implicitly encoded within joint generative or sequential modeling of observations and controls.

Joint video–action modeling for implicit inverse dynamics. A representative line of work, by coupling video and action generation for end-to-end learning, enables predicted visual dynamics to improve latent representations, where inverse dynamics is implicitly captured to generate actions [4]–[15]. A representative example is DreamZero [16], which enables leveraging large-scale action-free data without relying on repetitive demonstrations, leading to improved generalization across diverse tasks and environments [16]. Despite these advantages, using DreamZero as a representative example, we discuss three potential limitations.

Alignment. Such implicit formulations rely on the alignment between generated visual dynamics and actions, which is not explicitly enforced. As noted in the paper, *"we observe that the predicted actions closely align with the generated video"* [16], suggesting that such alignment may emerge implicitly without explicit constraints, though it is primarily supported by qualitative observations. Although joint modeling can produce seemingly consistent behavior, it does not guarantee that visual dynamics and actions are truly aligned. The model may effectively learn to predict visual transitions and actions independently, without enforcing precise correspondence between them. As a result, alignment is only implicitly learned and can be fragile, leading to misaligned behavior when predicted dynamics deviate from reality.

IDM may be less robust. By coupling video and action generation, these approaches learn predictive representations from visual dynamics, effectively mapping 2D image transitions to underlying 3D actions. This introduces an inherently under-constrained problem, as 2D observations do not uniquely determine the corresponding action. While models such as DreamZero leverage strong video priors to mitigate this ambiguity, the inverse mapping remains ill-posed, might make the learned dynamics less robust, especially under novel viewpoints or distribution shifts.

Lack of verification and learning from prediction errors. During inference, DreamZero replaces predicted frames with ground-truth observations to prevent error accumulation in closed-loop control [16]. While this stabilizes execution, it also removes opportunities for the model to verify and learn from its own prediction errors.

Sequence-based implicit inverse dynamics. Another line of work adopts sequence modeling, where observations, actions, and rewards are tokenized and jointly modeled via next-token prediction, as exemplified by iVideoGPT [17]. In such approaches, action information is entangled within the sequence and inferred implicitly through autoregressive generation.

B. Explicit Inverse Dynamics Modeling

In contrast to implicit formulations, explicit inverse dynamics models introduce a dedicated action prediction pathway, where actions are explicitly inferred from state transitions.

Two-frame / video-based IDM. A common formulation predicts actions directly from consecutive observations or planned visual goals. Early works learn to infer actions from video transitions or video-conditioned goals, e.g., predicting actions from pairs of frames or future visual plans [18]–[25]. These approaches treat inverse dynamics as an explicit mapping from visual transitions to actions, but often do not fully exploit the underlying physical structure of the environment. Moreover, inferring 3D actions from 2D image observations is inherently under-constrained, which may limit robustness, especially under novel viewpoints or distribution shifts.

Optical flow-based IDM. Another line of work leverages dense correspondences such as optical flow to bridge visual changes and actions. For example, actions can be inferred from estimated pixel correspondences or motion fields [26], which achieves zero-shot generalization on real-World scene, or through intermediate latent action representations derived from visual motion [27]. However, they may still suffer from information loss, leading to not precise action.

Tool-centric IDM. Tool-centric approaches instead focus on modeling the motion of manipulators(e.g., hand) or tools as a proxy for action. Methods such as large video planners and video-driven policy learning explicitly track tool

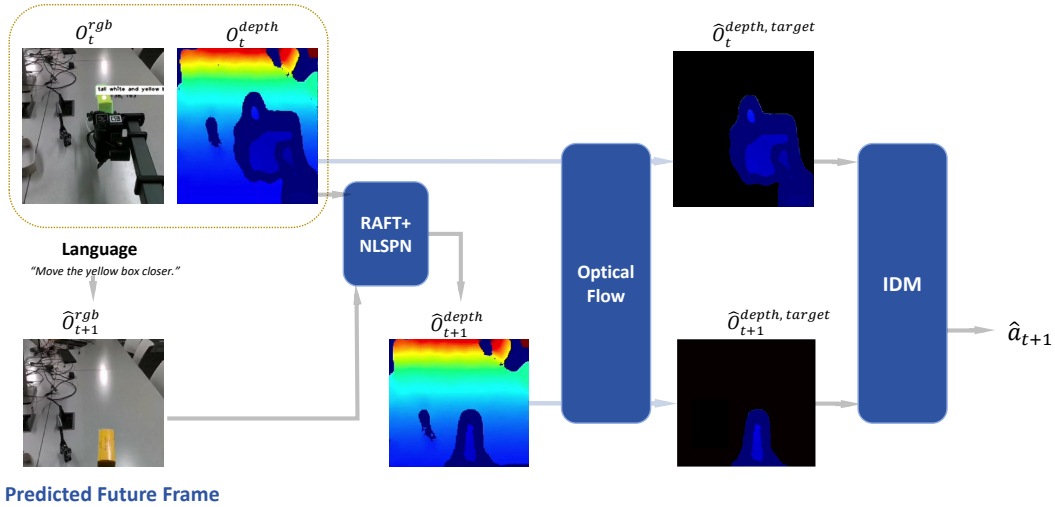

Fig. 1: Overview of the proposed IDM pipeline. Given the current RGB-D observation and a language instruction, a future frame is first predicted to represent the desired outcome. Depth is then propagated using optical flow (RAFT) and completed via NLSPN. Motion between frames is used to implicitly identify the manipulated object, yielding target-centric depth representations. Finally, an inverse dynamics model predicts the action that realizes the desired transition.

trajectories and map them to control signals [28], [29]. For example, Dreamitate [29] formulates policy learning as video generation followed by trajectory extraction and execution. Specifically, the model first generates a video of task execution conditioned on the current observation, from which a 2D trajectory of the tool is extracted. This trajectory is then lifted to 3D space by matching it to feasible trajectories via MegaPose, which is used to recover accurate end-effector states from RGB observations, then actions are obtained via inverse kinematics. The availability of 3D information (e.g., depth) enables accurate trajectory execution for fine-grained manipulation tasks. This also leads to strong generalization across unseen objects, environments, and embodiments, as the policy focuses on consistent tool motion.

However, such approaches, instead of learning the causal relationship between actions and object state transitions, they primarily track and execute tool trajectories (e.g., end-effector motion), making them closer to imitation learning of trajectories rather than true dynamics modeling. This introduces a structural mismatch: successful execution relies on the assumption that tool motion faithfully reflects object interaction, which may not always hold in complex environments. As a result, these methods may fail to capture the underlying action–object causality that is central to world modeling.

3D and object-centric IDM. More structured formulations further incorporate 3D information or object-centric representations. For instance, some methods recover 3D states or poses from visual observations and perform inverse kinematics to obtain actions [29], while others learn dense object descriptors or keypoint-based representations to explicitly construct inverse mappings [30], [31]. These approaches enables precise control and fine-grained manipulation. However, they often require additional assumptions on object structure or access to reliable 3D information.

Recent work reformulates robot manipulation as a 4D video generation problem instead of direct policy learning. *Geometry-aware 4D Video Generation for Robot Manipulation* proposes a multi-view diffusion framework that predicts future RGB frames together with per-pixel 3D pointmaps [32]. A key challenge in this setting is that videos from different viewpoints are inherently difficult to align due to occlusions, perspective changes, and the lack of explicit cross-view correspondences in image space. To address this, the method introduces cross-view attention and enforces geometric consistency via pointmap alignment, effectively lifting the alignment problem from 2D image space to a shared 3D representation. Nevertheless, the approach still relies on accurate geometry estimation and may degrade under severe occlusion or noisy depth prediction.

Latent representation IDM. Finally, recent works explore learning inverse dynamics in latent representation spaces, where actions are predicted from learned visual embeddings [33]–[36]. While scalable, such approaches depend on the quality of learned representations, typically produced by general-purpose encoders (e.g., ViT or CNN), which may lack precise spatial grounding for accurate control.

Overall, existing inverse dynamics models often exhibit limited robustness, partly due to insufficient spatial information for accurate action grounding.

C. Manipulation Based on 3D Data

Vision-based robotic manipulation methods can be broadly categorized according to their data representations and network architectures. **2D heatmap-based approaches** take depth images as input and employ convolutional neural networks (CNNs) to predict grasp quality scores in the form of 2D heatmaps, which are then projected into 3D space (e.g., Dex-Net 2.0, FC-GQ-CNN). These methods are simple, stable, and interpretable, and have been widely

adopted in industrial grasping. However, they rely heavily on camera models and only provide indirect 3D reasoning. In contrast, **keypoint or affordance-based methods** predict task-relevant keypoints or action regions from RGB or depth images using encoder architectures such as U-Net (e.g., Transporter Networks). These approaches are more data-efficient and generalize well across tasks, particularly in pick-and-place scenarios, but they typically suffer from limited spatial precision.

Moving beyond image-based representations, **point cloud-based methods** directly operate on 3D point sets and leverage architectures such as PointNet or Transformers to predict 6D object poses (e.g., PointNetGPD, Contact-GraspNet). By utilizing explicit geometric information, these methods achieve high accuracy and are well-suited for robotic manipulation, although they can be sensitive to sensor noise. **Voxel grid-based approaches** convert point clouds into structured 3D grids and apply 3D CNNs for dense prediction (e.g., VoxNet), offering regularized representations at the cost of high memory and computational demands. Finally, **SE(3)-equivariant methods** (e.g., SE(3)-Transformer, Tensor Field Networks) incorporate symmetry-aware modeling to ensure equivariance to 3D rotations and translations, leading to physically consistent representations and superior performance in 6D pose estimation. Despite their strong theoretical properties, these models are often computationally expensive and remain primarily in advanced research settings. Overall, these approaches reflect different trade-offs among accuracy, efficiency, and generalization, illustrating the evolution from indirect 2D perception to geometry-aware 3D understanding in robotic manipulation.

II. Potential Directions

Action–image consistency for joint video–action modeling. A potential direction is to improve alignment in joint video–action modeling for implicit inverse dynamics by explicitly enforcing consistency between predicted visual dynamics and generated actions.

Error-driven model verification and refinement. A potential direction is to leverage execution-time prediction errors as a learning signal to verify and update the model, rather than discarding them through ground-truth replacement.

Spatially grounded inverse dynamics for precise manipulation and improved generalization. We detail the proposed approach in the following section.

III. Method of the proposed IDM

We propose a **Spatially Grounded IDM** for **Precise Manipulation and Improved Generalization**(Fig 1).

Given the current observation and a language instruction, we first predict a future frame that reflects the desired outcome. Depth maps are then estimated for the predicted frame. We further extract target-centric depth using optical flow, where **the manipulated object is implicitly defined as regions exhibiting motion across frames**. Finally, an inverse dynamics model takes the resulting geometric transformation as input and predicts the action that realizes the transition.

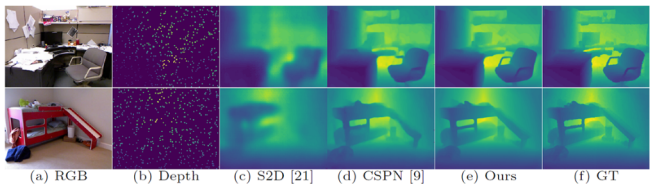

Fig. 2: Example of depth completion using NLSPN

A. Pipeline detail

Future Frame Prediction. We first predict a future observation $\hat{O}_{t+1}^{rgb}$ conditioned on the current observation O_t^{rgb} and the language instruction. This predicted frame represents the desired target state implied by the instruction.

Future Depth Estimation. Given the current RGB-D observation (O_t^{rgb}, O_t^{depth}) and the predicted future RGB frame $\hat{O}_{t+1}^{rgb}$, the goal is to estimate the corresponding future depth $\hat{O}_{t+1}^{depth}$. This can be achieved through several possible pipelines(Appendix A). Here, we adopt a flow-based depth propagation approach using RAFT for optical flow estimation and NLSPN for depth completion.

Specifically, we first estimate optical flow between O_t^{rgb} and $\hat{O}_{t+1}^{rgb}$ using RAFT to obtain pixel correspondences. The depth map O_t^{depth} is then warped to the target frame using the estimated flow, producing a sparse depth map $\tilde{O}_{t+1}^{depth}$. Due to occlusions and matching errors, the warped depth is incomplete. We then apply NLSPN to recover a dense depth map $\hat{O}_{t+1}^{depth}$ (Fig. 2).

Target Extraction via Motion. We extract target-centric depth using optical flow computed between O_t^{rgb} and $\hat{O}_{t+1}^{rgb}$. Specifically, the manipulated object is implicitly defined as regions exhibiting motion across frames. Applying this motion mask to the depth maps yields target-focused depth representations, denoted as $O_t^{depth,\text{target}}$ and $\hat{O}_{t+1}^{depth,\text{target}}$.

Inverse Dynamics Prediction. Finally, an inverse dynamics model takes the geometric transformation as input and predicts the action a_{t+1} that realizes the transition from $O_t^{depth,\text{target}}$ to $\hat{O}_{t+1}^{depth,\text{target}}$.

B. Advantages

Our pipeline offers several desirable properties. **Strong generalization through geometry.** By relying on spatially grounded geometric representations rather than appearance, the model naturally generalizes across variations in background, object appearance, and lighting conditions. Since the inverse dynamics model operates on target-centric depth, irrelevant visual details are abstracted away, retaining only essential geometric and spatial relationships. This also facilitates generalization across tasks that share similar subgoals, even when the overall task differs. **Accurate and fine-grained manipulation.** The use of explicit 3D information enables more precise spatial reasoning and supports fine

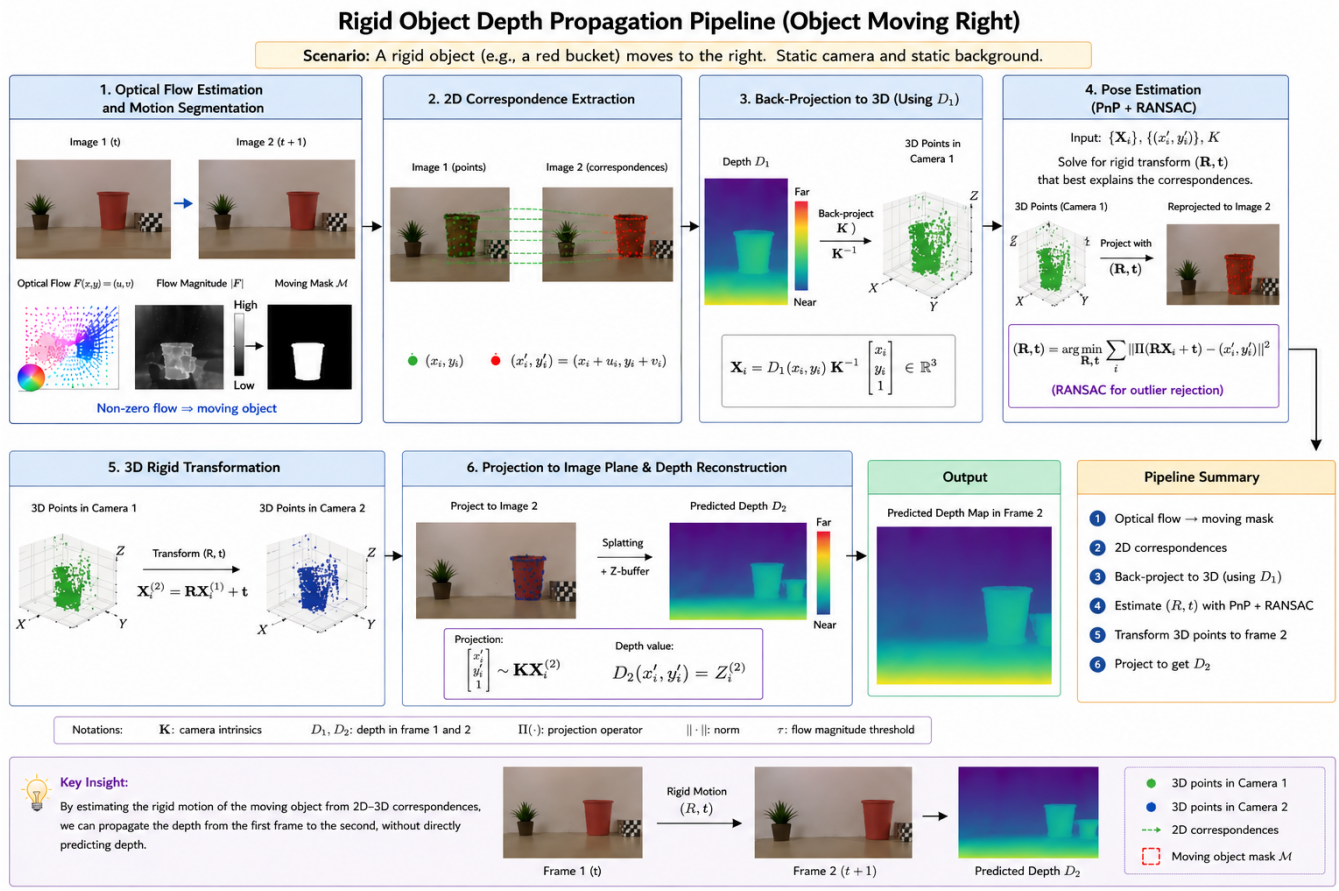

Fig. 3: Overview of the rigid object depth propagation pipeline. Given the depth map in the first frame, we first estimate optical flow and extract moving object regions via motion segmentation. The corresponding pixels are then back-projected to 3D using camera intrinsics, forming a point cloud. A rigid transformation is estimated via PnP with RANSAC using 2D–3D correspondences, and subsequently applied to propagate the 3D structure to the next frame. Finally, the transformed points are projected back to the image plane to reconstruct the depth map, with z-buffering used to handle occlusions. This pipeline enables depth estimation by recovering object motion rather than directly predicting depth.

manipulation. Compared to purely image-based representations, depth-based geometric cues improve robustness under viewpoint changes and allow accurate execution of detailed actions.

Implicit target identification via motion. The manipulated object is implicitly identified through motion, eliminating the need for explicit object detection or segmentation (e.g., SAM), and enabling a more flexible and scalable formulation.

Simplified and learnable inverse dynamics. The resulting inverse dynamics problem becomes significantly simpler, as the input captures only the essential geometric changes, without requiring the model to fit irrelevant visual details. This abstraction improves generalization while reducing model complexity, making it akin to classical control or inverse kinematics. As a result, the formulation enhances both learnability and data efficiency.

Appendix

Rigid Object Depth Propagation Pipeline

Problem Setup

Given:

- Two RGB images: I_1, I_2
- Depth map of the first frame: D_1
- Camera intrinsics: K
- Static camera and static background

Goal:

Estimate D_2 (or equivalently $\mathbf{X}_2$)

A. Geometry-Aware Depth Propagation with Motion Segmentation and PnP

Step 1: Optical Flow Estimation and Motion Segmentation: Compute optical flow:

$$F(x, y) = (u(x, y), v(x, y))$$

Generate moving object mask:

$$\mathcal{M} = (x, y) \mid |F(x, y)| > \tau$$

Optional refinement:

- Median filtering
- Connected component filtering

Step 2: 2D Correspondence Extraction: For pixels in $\mathcal{M}$:

$$(x_i, y_i) \to (x_i', y_i') = (x_i + u_i, ; y_i + v_i)$$

Step 3: Back-Projection to 3D (Point Cloud Construction):

$$\mathbf{X}_i = D_1(x_i, y_i); K^{-1} \begin{bmatrix} x_i & y_i & 1 \end{bmatrix}$$

Result:

$$\mathbf{X}_i \subset \mathbb{R}^3$$

Step 4: Pose Estimation via PnP with RANSAC:

$$(R, t) = \arg\min_{R,t} \sum_i |\Pi(R\mathbf{X}_i + t) - (x_i', y_i')|^2$$

Using:

- Perspective-n-Point (PnP)
- RANSAC for outlier rejection

Step 5: 3D Rigid Transformation

$$\mathbf{X}_i^{(2)} = R\mathbf{X}_i^{(1)} + t$$

Step 6: Projection to Image Plane (Depth Reconstruction):

$$\begin{bmatrix} x_i' & y_i' & 1 \end{bmatrix} \sim K\mathbf{X}_i^{(2)}$$

Depth value:

$$D_2(x_i', y_i') = Z_i^{(2)}$$

Implementation Notes:

- Use forward splatting for projection
- Apply z-buffer to resolve occlusions:

$$D_2(x, y) = \min Z \text{ at each pixel}$$

Robustness Enhancements

- Weighted optimization:

$$\min \sum_i w_i \cdot |e_i|$$

- Robust loss (Huber / Cauchy)
- Flow confidence filtering

Summary

$$\mathbf{X}_2 = R \cdot \Pi^{-1}(D_1) + t \quad \Rightarrow \quad D_2 = \Pi(\mathbf{X}_2)$$

Key Insight:

Depth propagation is achieved by recovering rigid motion rather than directly predicting depth.

B. End-to-end Pipeline: RAFT-3D

Inputs:

$$I_1, I_2 \in \mathbb{R}^{H \times W \times 3}$$
$$D_1 \in \mathbb{R}^{H \times W}$$
$$K \in \mathbb{R}^{3 \times 3}$$

Estimate dense 3D motion (scene flow) ΔX for each pixel. But its generalization is limited.

References

[1] Q. Garrido, M. Assran, N. Ballas, A. Bardes, L. Najman, and Y. LeCun, "Learning and leveraging world models in visual representation learning," 2024. [Online]. Available: https://arxiv.org/abs/2403.00504

[2] B. Chen, T. Zhang, H. Geng, K. Song, C. Zhang, P. Li, W. T. Freeman, J. Malik, P. Abbeel, R. Tedrake, V. Sitzmann, and Y. Du, "Large video planner enables generalizable robot control," 2025. [Online]. Available: https://arxiv.org/abs/2512.15840

[3] Y. Liu, F. Feng, L. Kong, W. Lu, J. Tang, K. Zhang, K. Murphy, C. Finn, and Y. Du, "World action verifier: Self-improving world models via forward-inverse asymmetry," *arXiv preprint arXiv:2604.01985*, 2026.

[4] C. Zhu, R. Yu, S. Feng, B. Burchfiel, P. Shah, and A. Gupta, "Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets," *arXiv preprint arXiv:2504.02792*, 2025.

[5] H. et al., "Video prediction policy: A generalist robot policy with predictive visual representations," 2024.

[6] K. et al., "Cosmos policy: Fine-tuning video models for visuomotor control and planning," 2026.

[7] L. et al., "Video generators are robot policies," 2025.

[8] ——, "Genie envisioner: A unified world foundation platform for robotic manipulation," 2025.

[9] P. et al., "Mimicvideo: Video-action models for generalizable robot control beyond vlas," 2025.

[10] C.-L. Cheang, G. Chen, Y. Jing, T. Kong, H. Li, Y. Li, Y. Liu, H. Wu, J. Xu, Y. Yang *et al.*, "Gr-2: A generative video-language-action model with web-scale knowledge for robot manipulation," *arXiv preprint arXiv:2410.06158*, 2024.

[11] L. et al., "Unified video action model," 2025.

[12] W. et al., "Dual-stream diffusion for world-model augmented vision-language-action model," 2025.

[13] ——, "Unleashing large-scale video generative pre-training for visual robot manipulation," 2024.

[14] Z. et al., "Cot-vla: Visual chain-of-thought reasoning for vision-language-action models," 2025.

[15] ——, "Tracevla: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies," 2025.

[16] S. Ye, Y. Ge, K. Zheng, S. Gao, S. Yu, G. Kurian, S. Indupuru, Y. L. Tan, C. Zhu, J. Xiang, A. Malik, K. Lee, W. Liang, N. Ranawaka, J. Gu, Y. Xu, G. Wang, F. Hu, A. Narayan, J. Bjorck, J. Wang, G. Kim, D. Niu, R. Zheng, Y. Xie, J. Wu, Q. Wang, R. Julian, D. Xu, Y. Du, Y. Chebotar, S. Reed, J. Kautz, Y. Zhu, L. J. Fan, and J. Jang, "World action models are zero-shot policies," 2026. [Online]. Available: https://arxiv.org/abs/2602.15922

[17] J. Wu, S. Yin, N. Feng, X. He, D. Li, J. Hao, and M. Long, "ivideogpt: Interactive videogpts are scalable world models," *Advances in Neural Information Processing Systems*, vol. 37, pp. 68 082–68 119, 2024.

[18] Y. Du, S. Yang, B. Dai, H. Dai, O. Nachum, J. Tenenbaum, D. Schuurmans, and P. Abbeel, "Learning universal policies via text-guided video generation," *Advances in neural information processing systems*, vol. 36, pp. 9156–9172, 2023.

[19] B. Baker, I. Akkaya, P. Zhokov, J. Huizinga, J. Tang, A. Ecoffet, B. Houghton, R. Sampedro, and J. Clune, "Video pretraining (vpt): Learning to act by watching unlabeled online videos," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 639–24 654, 2022.

[20] S. Zhou, Y. Du, J. Chen, Y. Li, D.-Y. Yeung, and C. Gan, "Robodreamer: Learning compositional world models for robot imagination," *arXiv preprint arXiv:2404.12377*, 2024.

[21] H. Bharadhwaj, D. Dwibedi, A. Gupta, S. Tulsiani, C. Doersch, T. Xiao, D. Shah, F. Xia, D. Sadigh, and S. Kirmani, "Gen2act: Human video generation in novel scenarios enables generalizable robot manipulation," *arXiv preprint arXiv:2409.16283*, 2024.

[22] J. Jang, S. Ye, Z. Lin, J. Xiang, J. Bjorck, Y. Fang, F. Hu, S. Huang, K. Kundalia, Y.-C. Lin *et al.*, "Dreamgen: Unlocking generalization in robot learning through video world models," *arXiv preprint arXiv:2505.12705*, 2025.

[23] K. Black, M. Nakamoto, P. Atreya, H. Walke, C. Finn, A. Kumar, and S. Levine, "Zero-shot robotic manipulation with pretrained image-editing diffusion models," *arXiv preprint arXiv:2310.10639*, 2023.

[24] J. Cai, Z. Cai, J. Cao, Y. Chen, Z. He, L. Jiang, H. Li, H. Li, Y. Li, Y. Liu *et al.*, "Internvla-a1: Unifying understanding, generation and action for robotic manipulation," *arXiv preprint arXiv:2601.02456*, 2026.

[25] X. Chen, J. Guo, T. He, C. Zhang, P. Zhang, D. C. Yang, L. Zhao, and J. Bian, "Igor: Image-goal representations are the atomic building blocks for next-level generalization in embodied ai."

[26] P.-C. Ko, J. Mao, Y. Du, S.-H. Sun, and J. B. Tenenbaum, "Learning to act from actionless videos through dense correspondences," *arXiv preprint arXiv:2310.08576*, 2023.

[27] H. Bi, H. Tan, S. Xie, Z. Wang, S. Huang, H. Liu, R. Zhao, Y. Feng, C. Xiang, Y. Rong *et al.*, "Motus: A unified latent action world model," *arXiv preprint arXiv:2512.13030*, 2025.

[28] B. Chen, T. Zhang, H. Geng, K. Song, C. Zhang, P. Li, W. T. Freeman, J. Malik, P. Abbeel, R. Tedrake *et al.*, "Large video planner enables generalizable robot control," *arXiv preprint arXiv:2512.15840*, 2025.

[29] J. Liang, R. Liu, E. Ozguroglu, S. Sudhakar, A. Dave, P. Tokmakov, S. Song, and C. Vondrick, "Dreamitate: Real-world visuomotor policy learning via video generation," *arXiv preprint arXiv:2406.16862*, 2024.

[30] P. R. Florence, L. Manuelli, and R. Tedrake, "Dense object nets: Learning dense visual object descriptors by and for robotic manipulation," *arXiv preprint arXiv:1806.08756*, 2018.

[31] A. Zeng, P. Florence, J. Tompson, S. Welker, J. Chien, M. Attarian, T. Armstrong, I. Krasin, D. Duong, V. Sindhwani *et al.*, "Transporter networks: Rearranging the visual world for robotic manipulation," in *Conference on Robot Learning*. PMLR, 2021, pp. 726–747.

[32] Z. Liu, S. Li, E. Cousineau, S. Feng, B. Burchfiel, and S. Song, "Geometry-aware 4d video generation for robot manipulation," 2026. [Online]. Available: https://arxiv.org/abs/2507.01099

[33] Y. Hu, Y. Guo, P. Wang, X. Chen, Y.-J. Wang, J. Zhang, K. Sreenath, C. Lu, and J. Chen, "Video prediction policy: A generalist robot policy with predictive visual representations," *arXiv preprint arXiv:2412.14803*, 2024.

[34] C. Luo, Z. Zeng, M. Jia, Y. Du, and C. Sun, "Self-adapting improvement loops for robotic learning," *arXiv preprint arXiv:2506.06658*, 2025.

[35] Q. Lv, W. Kong, H. Li, J. Zeng, Z. Qiu, D. Qu, H. Song, Q. Chen, X. Deng, and J. Pang, "F1: A vision-language-action model bridging understanding and generation to actions," *arXiv preprint arXiv:2509.06951*, 2025.

[36] Y. Tian, S. Yang, J. Zeng, P. Wang, D. Lin, H. Dong, and J. Pang, "Predictive inverse dynamics models are scalable learners for robotic manipulation," *arXiv preprint arXiv:2412.15109*, 2024.